LF3 Worked Examples: Graphing Lines

Recording companion for the LF3 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What two points can I plot? Every line needs two points. Intercepts give points with a 0 in them: an x-intercept of 3 is the point (3, 0). A slope gives a second point from the first: a slope of −2/3 means right 3, down 2. From an equation, find a point and the slope, or both intercepts. $x = c$ is a vertical line and $y = c$ is a horizontal one.

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Example 1: Two Intercepts

$$x\text{−intercept } -3$$
$$y\text{−intercept } 5$$

Question

Which two points do you plot to graph this line?

- **A.** (0, −3) and (5, 0)
- **B.** (−3, 0) and (0, 5)
- **C.** (−3, 5) only
- **D.** (3, 0) and (0, 5)

Answer and reasoning

An intercept is where the line crosses an axis. The x-intercept is on the x-axis, where $y=0$: $(-3,0)$. The y-intercept is on the y-axis, where $x=0$: $(0,5)$. Plot both and draw the line through them. $(0,-3)$ and $(5,0)$ put each 0 in the wrong place; $(-3,5)$ is a single point, which is not enough for a line and is not on this one; and $(3,0)$ drops the minus sign. ***(−3,0)* and *(0,5)*.**

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Example 2: Point and Intercept

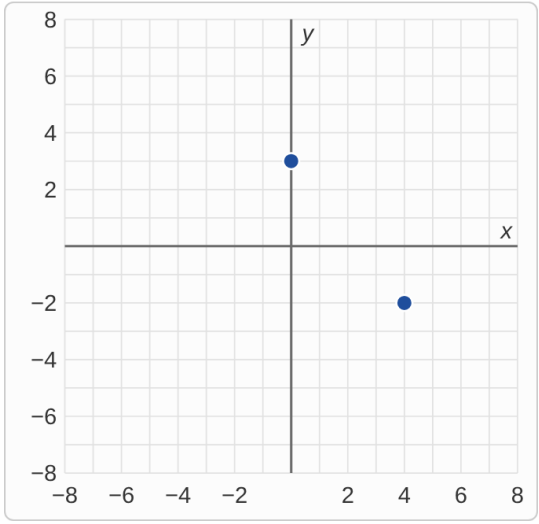


Figure: A grid from −8 to 8 in both directions, with grid lines every 1 unit. Two points are plotted: (0, 3) on the y-axis and (4, −2). No line is drawn yet.

$$(4,-2), \quad y\text{−intercept } 3$$

Question

What is the slope of the line through these two points?

- **A.** $-5/4$
- **B.** $5/4$
- **C.** $-4/5$
- **D.** $-1/4$

Answer and reasoning

The y-intercept 3 is the point $(0,3)$. From $(0,3)$ to $(4,-2)$: over 4, down 5, so $m=(-2-3)/(4-0)=-5/4$. You do not need the slope to draw this line, since two points are enough, but it tells you which way the line goes. $5/4$ misses that the line goes down; $-4/5$ is run over rise; and $-1/4$ works out $-2-3$ as -1. ***−5/4*.**

Example 3: Slope and Intercept

$$m=-2, \quad x\text{-intercept } 3$$

Question

You plot (3, 0). Which point should you plot next?

A. (4, 2)
B. (5, −1)
C. (3, −2)
D. (4, −2)

Answer and reasoning

The *x*-intercept 3 is the point *(3,0)*. A slope of *−2* is *(−2)/1*: right 1, down 2, to *(4,−2)*. Going left 1 and up 2, to *(2,2)*, works too. *(4,2)* goes up instead of down; *(5,−1)* moves right 2 and down 1, which is slope *−1/2*; and *(3,−2)* moves down without moving across, which does not use a run at all. ***(4,−2)*.**

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Example 4: Point and Slope

$$m=3, \quad (-2,1)$$

Question

You plot (−2, 1). Which point is also on the line?

- **A.** (−3, 4)
- **B.** (−1, 4)
- **C.** (1, 2)
- **D.** (−2, 4)

Answer and reasoning

Slope 3 means *3/1*: right 1, up 3. From *(−2,1)* that lands on *(−1,4)*. *(−3,4)* goes left 1 and up 3, which is slope *−3*; *(1,2)* goes right 3 and up 1, slope *1/3*; and *(−2,4)* goes straight up, with no run. ***(−1,4).***

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Example 5: Slope-Intercept Form

$$y=-2/3x+4$$

Question

How do you graph this line?

A. Start at (4, 0); go right 3, down 2.
B. Start at (0, 4); go right 2, down 3.
C. Start at (0, 4); go right 3, down 2.
D. Start at (0, 4); go right 3, up 2.

Answer and reasoning

In $y=mx+b$, $b=4$ is the y-intercept, the point $(0,4)$. The slope $-2/3$ is rise over run: down 2 for every 3 across. So from $(0,4)$ go right 3 and down 2, to $(3,2)$. $(4,0)$ puts the intercept on the wrong axis; right 2, down 3 flips rise and run; and up 2 misses the minus sign. **Start at $(0,4)$; go right 3, down 2.**

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Example 6: Point-Slope Form

$$y-2=-3(x+1)$$

Question

Which point and slope does this equation give?

A. The point (−1, 2), slope −3
B. The point (1, −2), slope −3
C. The point (−1, 2), slope 3
D. The point (2, −1), slope −3

Answer and reasoning

Point-slope form is $y-y_1=m(x-x_1)$. Here $y-2$ means $y_1=2$, and $x+1=x-(-1)$ means $x_1=-1$. The slope is the number in front, -3. Plot $(-1,2)$, then go right 1 and down 3. $(1,-2)$ reads the signs as written instead of flipping them; slope 3 drops the minus; and $(2,-1)$ swaps the coordinates. ***(−1,2)*, slope *−3*.**

Example 7: Standard Form 1

$$2x-3y=6$$

Question

Which two intercepts can you plot to graph this line?

A. (3, 0) and (0, 2)
B. (3, 0) and (0, −2)
C. (−3, 0) and (0, −2)
D. (2, 0) and (0, −3)

Answer and reasoning

Set $y=0$: $2x=6$, so $x=3$, the point $(3,0)$. Set $x=0$: $-3y=6$, so $y=-2$, the point $(0,-2)$. Standard form makes intercepts quick. $(0,2)$ loses the sign from dividing by -3; $(-3,0)$ gives x a sign it does not have; and $(2,0)$ and $(0,-3)$ divide 6 by the other variable's number. ***(3,0)* and *(0,−2)*.**

Example 8: Standard Form 2

$$3x+4y=8$$

Question

The x-intercept here is 8/3, which is hard to plot. Which two points work better?

A. (0, 2) and (4, −1)
B. (0, 2) and (4, 5)
C. (0, 8) and (4, 5)
D. (0, 2) and (3, −2)

Answer and reasoning

Solve for y: $4y=-3x+8$, so $y=-3/4x+2$. Start at the y-intercept $(0,2)$ and use the slope: right 4, down 3, to $(4,-1)$. Check: $3(4)+4(-1)=8$. This is the move to make whenever a standard-form intercept is a fraction. $(4,5)$ goes up instead of down; $(0,8)$ forgets to divide by 4; and $(3,-2)$ goes right 3 and down 4, which flips rise and run. ***(0,2)* and *(4,−1)*.**

Example 9: x = c

$$x=-4$$

Question

What does the graph of this line look like?

A. A vertical line through (−4, 0)
B. A horizontal line through (0, −4)
C. A line through the origin with slope −4
D. A vertical line through (4, 0)

Answer and reasoning

The equation says every point has $x=-4$, whatever y is: $(-4,0)$, $(-4,3)$, $(-4,-7)$. Those points stack straight up and down, so the graph is the vertical line through $(-4,0)$. A horizontal line through $(0,-4)$ is $y=-4$; slope -4 through the origin is $y=-4x$; and $(4,0)$ drops the sign. **A vertical line through $(-4,0)$.**

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Example 10: Name the Line

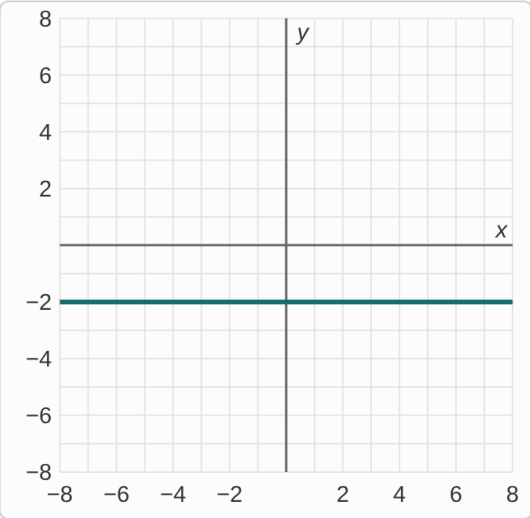


Figure: A straight line on a grid from −8 to 8 in both directions, with grid lines every 1 unit. It runs straight across at height −2, through (−8, −2), (0, −2) and (8, −2).

Question

Which equation does this graph show?

- **A.** $x=-2$
- **B.** $y=-2x$
- **C.** $y=-2$
- **D.** $y=x-2$

Answer and reasoning

Every point on this line has height -2, whatever its x: $(-8,-2)$, $(0,-2)$, $(8,-2)$. That is the equation $y=-2$. $x=-2$ would be an up-and-down line through $(-2,0)$; $y=-2x$ goes through the origin and falls steeply; and $y=x-2$ rises with slope 1. ***y=−2*.**

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Example 11: Slope 0

$$m=0, \quad (2,-5)$$

Question

Which is the equation of this line?

A. $x=2$
B. $y=-5$
C. $y=2$
D. $y=-5x$

Answer and reasoning

Slope 0 means the height never changes. The line goes through $(2,-5)$, so its height is always -5: $y=-5$. Point-slope form agrees: $y+5=0(x-2)$, so $y=-5$. $x=2$ is the up-and-down line, whose slope is undefined, not 0; $y=2$ uses the wrong coordinate; and $y=-5x$ treats the -5 as a slope. ***y=−5*.**

Example 12: Undefined Slope

$$m=(y_2-y_1)/(x_2-x_1)$$

Question

Which of these lines has an undefined slope?

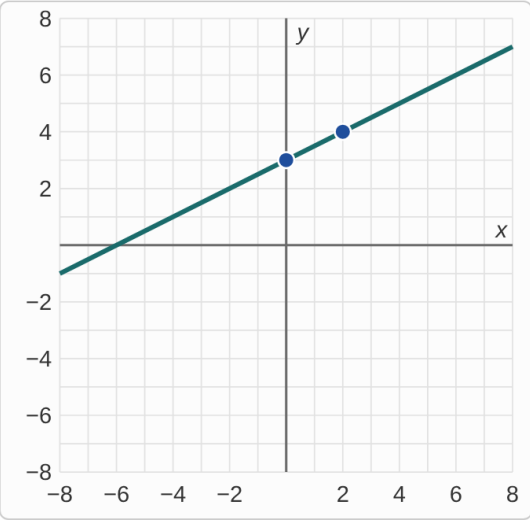
- **A.** $y=6$
- **B.** $x=6$
- **C.** $y=6x$
- **D.** $y=x+6$

Answer and reasoning

$x=6$ is the vertical line through $(6,0)$. Between any two of its points the run is 0, and dividing by 0 is undefined. $y=6$ is horizontal, with slope 0, not undefined; $y=6x$ has slope 6; and $y=x+6$ has slope 1. ***x=6*.**

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Example 13: Slip 1



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (0, 3) and (2, 4) and drew the straight line through them, which goes up from left to right.

$$y=-1/2x+3$$

Question

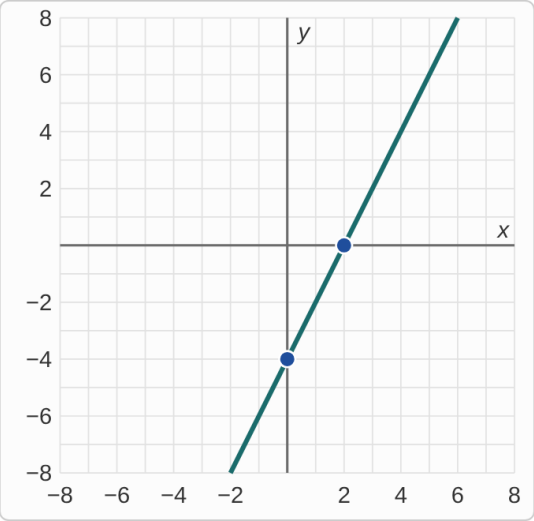
A student graphed this line. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** It should go down: right 2, down 1 from (0, 3).
- **C.** The y-intercept should be the point (3, 0) instead.
- **D.** Slope 1/2 means right 1, up 2, starting from (0, 3).

Answer and reasoning

The starting point *(0,3)* is right. But the slope is *−1/2*, so from *(0,3)* the line goes right 2 and down 1, to *(2,2)*. The student went up, to *(2,4)*, which is slope *+1/2*. The *y*-intercept is on the *y*-axis, *(0,3)*, not *(3,0)*; and right 1, up 2 would be slope 2. **From *(0,3)*, go right 2 and down 1.**

Example 14: Slip 2



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (2, 0) and (0, −4) and drew the straight line through them.

$$x\text{−intercept } 2$$
$$y\text{−intercept } -4$$

Question

A student graphed the line with these intercepts. What is the mistake, if any?

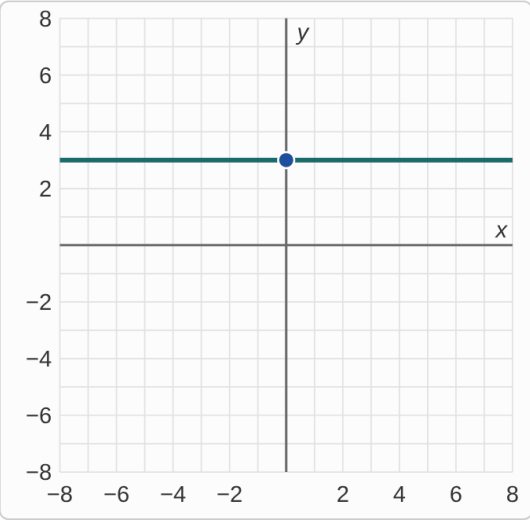
- **A.** There is no mistake.
- **B.** The points should be (0, 2) and (−4, 0).
- **C.** The line should go down from left to right.
- **D.** A line needs a slope, not two intercepts.

Answer and reasoning

The x-intercept 2 is $(2,0)$ and the y-intercept -4 is $(0,-4)$, and the student plotted both and drew the line through them. The student is right: the line is $y=2x-4$. Swapping the coordinates puts each point on the wrong axis; this line rises, because from $(0,-4)$ to

(2,0) it goes up 4 and over 2; and two points are always enough for a line. **There is no mistake.**

Example 15: Slip 3



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (0, 3) and drew a straight line through it that runs left to right at height 3.

$$x=3$$

Question

A student graphed this line. What is the mistake, if any?

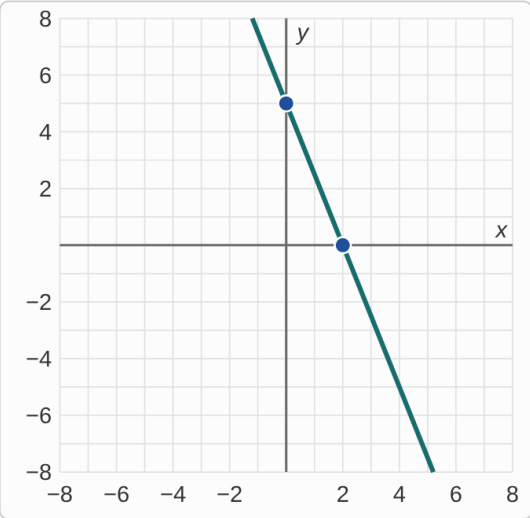
- **A.** There is no mistake.
- **B.** It should be the line through the origin with slope 3.
- **C.** x = 3 is the up-and-down line through (3, 0).
- **D.** It should be the left-to-right line at height −3.

Answer and reasoning

$x=3$ says every point has x-coordinate 3: $(3,0)$, $(3,5)$, $(3,-2)$. Those points line up vertically, through $(3,0)$. The student drew $y=3$, the horizontal line. The line through the origin with slope 3 is $y=3x$; and height -3 is $y=-3$, a horizontal line again. **Draw the vertical line through $(3,0)$.**

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Example 16: Slip 4



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (2, 0) and (0, 5) and drew the straight line through them.

$$2x+5y=10$$

Question

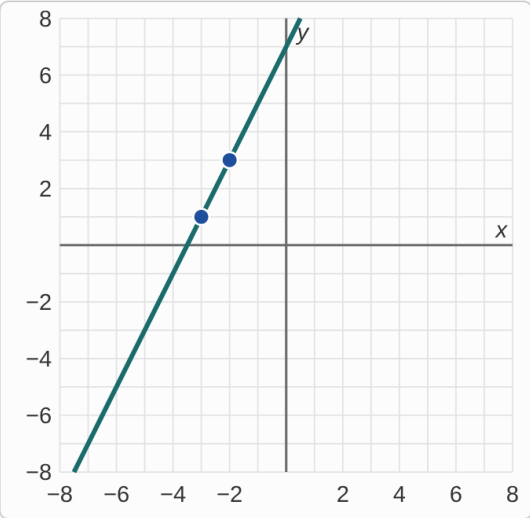
A student graphed this line using its intercepts. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** The line should go up, since both numbers are positive.
- **C.** The intercepts should be (−5, 0) and (0, −2).
- **D.** The intercepts are swapped: they are (5, 0) and (0, 2).

Answer and reasoning

Set $y=0$: $2x=10$, so $x=5$, the point $(5,0)$. Set $x=0$: $5y=10$, so $y=2$, the point $(0,2)$. The student divided 10 by the wrong number each time. Check the student's point: $2(2)+5(0)=4$, not 10. Positive A and B make the line go down, since the slope is $-2/5$; and nothing here makes an intercept negative. **Plot $(5,0)$ and $(0,2)$.**

Example 17: Slip 5



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (−3, 1) and (−2, 3) and drew the straight line through them.

$$y+1=2(x-3)$$

Question

A student graphed this line. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** The point is (3, −1); the student flipped both signs.
- **C.** The slope should be 1/2, so right 2 and up 1.
- **D.** The point should be (1, −3), with the numbers swapped.

Answer and reasoning

Point-slope form subtracts the coordinates: $y+1=y-(-1)$, so $y_1=-1$, and $x-3$ means $x_1=3$. The point is $(3,-1)$. The student used $(-3,1)$. The slope 2 was used correctly: right 1, up 2. Slope $1/2$ would flip rise and run; and swapping the coordinates gives a point that is not on the line. **Plot $(3,-1)$, then right 1 and up 2.**

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